REG: 222009343

YEAR III Computer and Software Engineering

Written part

(1) Dependencies I had used and why are:

* I used http package (http: ^1.2.0): so I chose this package because it provides simple and ready-made methods for making HTTP requests such as http.get(), http.post(), http.put() and http.delete() - these methods match exactly the CRUD operations needed to communicate with the JSONPlaceholder REST API. It is the official and most commonly used package for API consumption in flutter.

(2) How I handle API exceptions: API exceptions are handled using try/catch blocks around every network call. There are three levels of error handling:

* Network errors: Caught using http.ClientException which fires when there is no Internet connection or the server is unreachable. The user sees a message like "please check your Internet connection"
* Server errors: After successful connection we check response.statusCode. If it is not 200 or 201 we throw a custom Exception with the status code.
* Unexpected errors:
* Missing / invalid fields: If the JSON is missing expected fields like id, title or body the Post.fromJson() factory constructor will throw an error which is also caught by the catch(e) block.

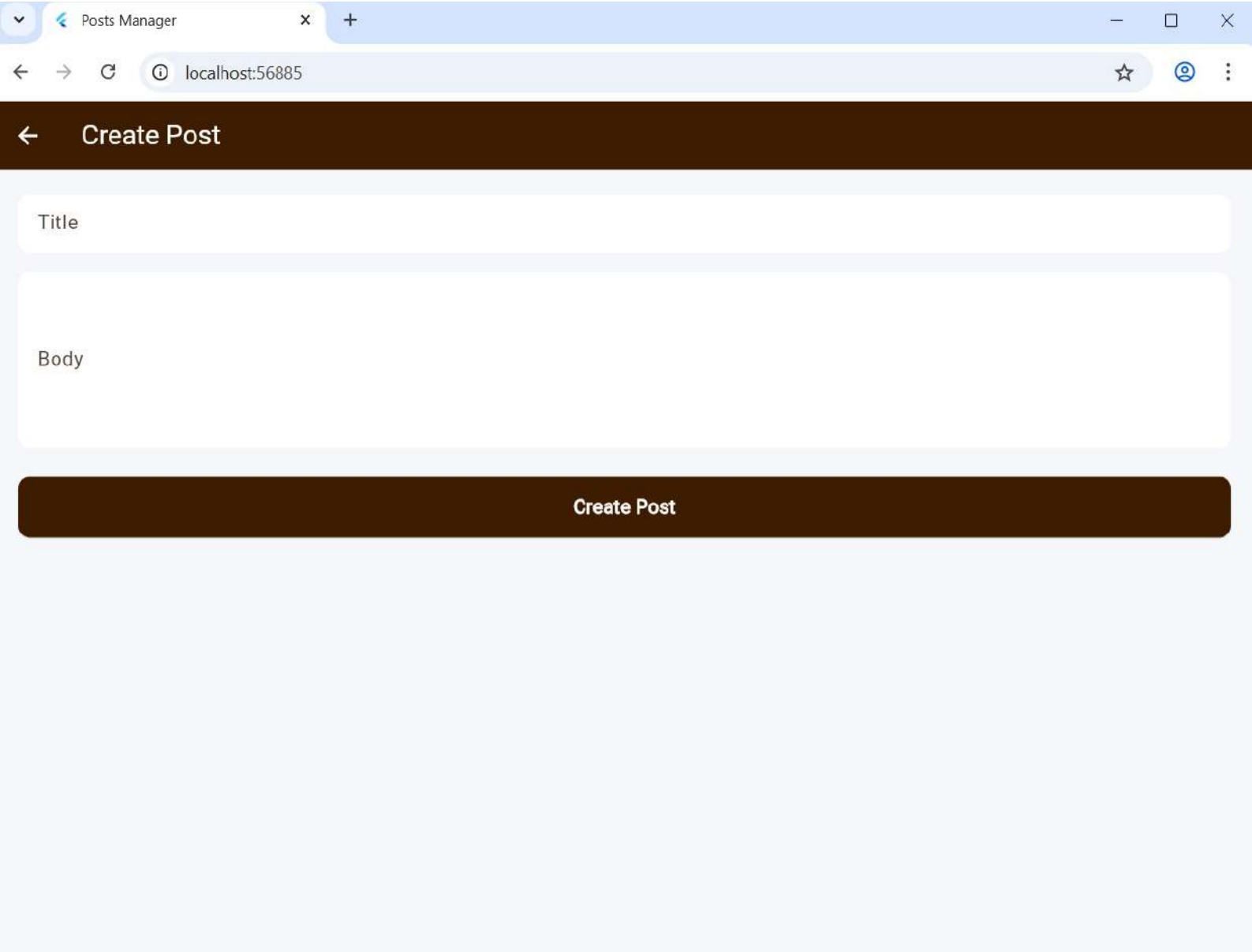
(3) FUTURE Builder and UI depends on it:

FutureBuilder is a flutter widget that listens to a future and automatically rebuilds the UI whenever the future's state changes. It takes two parameters: future, which is the async API call and builder which returns different widgets based on the asyncSnapshot state.

In my app we used FutureBuilder<List<Post>> in the PostListScreen. It handles three states:

- Connection state. waiting → displays a CircularProgressIndicator (loading spinner) while the API call is in progress
- Snapshot.hasError → displays a red error message with a Retry button if the API called failed
- Snapshot.hasData → displays the ListView.builder with all posts when data is successfully returned.

The entire body of PostsListScreen depends on the FutureBuilder, nothing is shown to the user until the future resolves. The future is stored in _postsFuture and is refreshed by calling setState() inside the _refresh() method


Posts Manager
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Create Post
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Create Post

Posts Manager

localhost:56885

Posts Manager

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Posts Manager

localhost:56885

Post Details

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Posts Manager

localhost:56885

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Save Changes

Posts Manager

localhost:56885

Posts Manager

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